Performance Evaluation of Lightweight LLMs Comparing on Colab and Edge Devices

Cross-Platform Analysis: Colab T4 GPU vs. NVIDIA Jetson Xavier NX

Neural Network and Deep Learning - EE608T

Under Guidance of
Dr. Rahul Jashvantbhai Pandya
Associate Professor, Electrical, Electronics and Communication Engineering (EECE)
Indian Institute of Technology, Dharwad

Presented by
Arjun Ambekar
EE25DP007
PhD Scholar, Department of Electrical, Electronics and Communication Engineering (EECE)
Indian Institute of Technology, Dharwad

Outline

1. Introduction
2. Methodology
3. Colab T4 Results
4. Jetson Xavier NX Results
5. Cross-Platform Comparison
6. Thermal Analysis
7. Analysis
8. Conclusion

Why Edge LLM Deployment?

The Problem

- Cloud inference: high latency, cost, privacy risk
- Edge devices: bounded memory, thermal limits, strict energy budgets
- Sub-2B models now rival 7B models from 2023

Our Questions

1. Which ultra-light model has the best speed–quality–energy trade-off on Jetson?
2. How does edge performance differ from cloud?
3. Can Xavier NX sustain inference under thermal load?

Models Under Test

Model	Params	Context	Disk	Architecture
Gemma 3 1B-IT	1.00 B	32 K	2.03 GB	SWA 5:1, SentencePiece 262K
Qwen 3 0.6B	0.75 B	32 K	1.52 GB	GQA + SwiGLU + RoPE
Qwen 3 1.7B	1.72 B	32 K	4.08 GB	Same family, $3\times$ scale

- **Gemma 3** (Google): sliding-window attention, text-only at 1B
- **Qwen 3** (Alibaba): hybrid thinking/non-thinking modes, Apache 2.0
- All: open-weight, instruction-tuned, fp16 inference

Hardware Platforms

	Colab T4	Jetson Xavier NX
GPU	NVIDIA T4 (2560 cores, Turing)	Volta integrated (384 cores)
Memory	15 GB GDDR6 (320 GB/s)	16 GB LPDDR4x (51.2 GB/s)
CPU	x86_64 Intel Xeon	ARMv8.2 Carmel 6-core
Power	∼70 W GPU TDP	∼15 W MAXN mode
Storage	SSD	eMMC
CUDA	12.8	11.4 (JetPack 5.1.2)

Key bandwidth gap

Xavier NX: 51.2 GB/s vs T4: 320 GB/s $\Rightarrow$ **6.25× less bandwidth**

Experimental Controls

Common Config

- Precision: `float16`
- Attention: `SDPA`
- Decoding: greedy (`do_sample=False`)
- Max tokens: 256
- Seed: 42

Controls

- 3 warmup runs discarded
- 3 measurement runs, median reported
- CUDA `synchronize()` before/after
- Xavier: `nvpmodel -m 0` + `jetson_clocks`
- Power via `tegrastats` @ 200 ms

Experimental Controls

5 prompts: short QA, instruction, reasoning, summarization, code generation

- **Short QA**: "What is the capital of Australia? Answer in one sentence."
- **Instruction**: "List three tips for writing clear technical documentation."
- **Reasoning**: "If a train leaves at 9:15 and arrives at 11:42, how long was the trip? Show your steps."
- **Summarization**: "Summarize in two sentences: The Industrial Revolution began in Britain in the late 18th century and spread through Europe and North America. It shifted economies from agrarian to manufacturing-based, introduced steam power and mechanized textile production, and drove mass urbanization. Working conditions were often poor, prompting later labor reforms."
- **Code**: "Write a Python function that returns the nth Fibonacci number iteratively."

Metrics Collected

Metric	What it measures	How
TTFT (ms)	Responsiveness	First token from `TextIteratorStreamer`
Decode (tok/s)	Generation speed	$(n_{out} - 1)/(t_{end} - t_{first})$
Peak VRAM (GB)	Memory footprint	`max_memory_allocated()`
Perplexity	Quality proxy	WikiText-2 sliding window
Power (W)	Energy draw	`tegrastats` GPU+CPU rails
Energy (J/tok)	Efficiency	Power $\times$ time / tokens

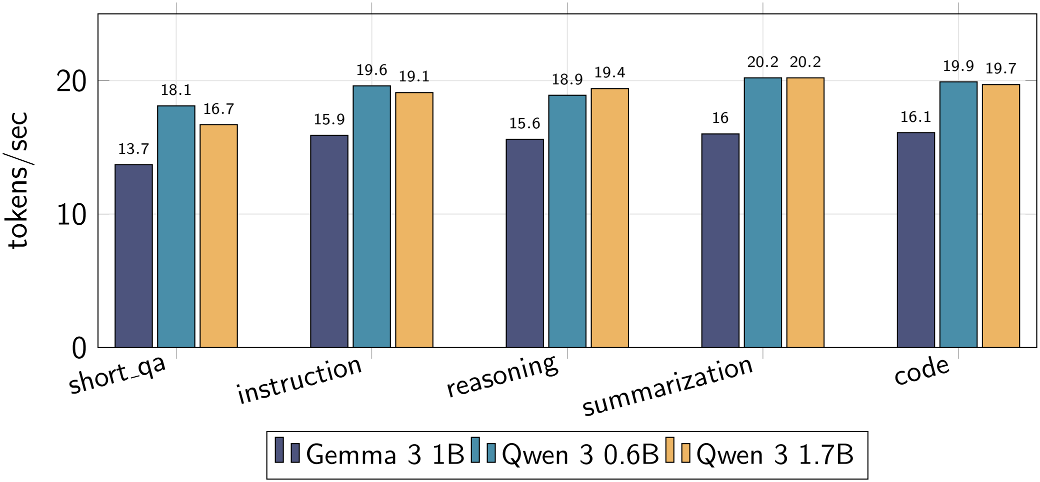
Colab T4 — Summary (Thinking OFF)

Model	TTFT (ms)	tok/s	VRAM (GB)	Disk (GB)	PPL ↓
Gemma 3 1B	67.4	15.9	2.02	2.03	26.57
Qwen 3 0.6B	54.7	**19.6**	**1.24**	**1.52**	21.23
Qwen 3 1.7B	55.2	19.4	3.50	4.08	**17.02**

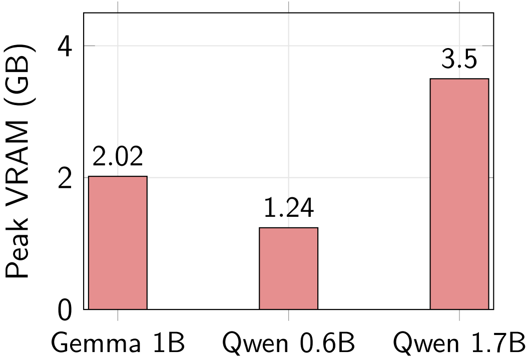
Key finding

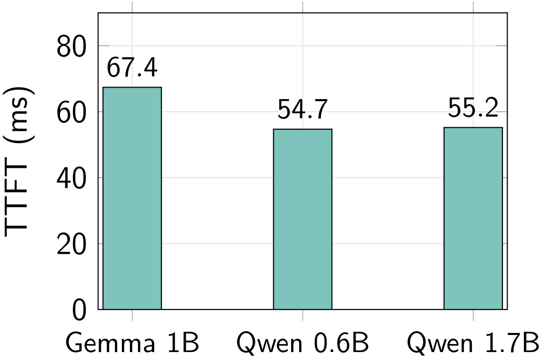
Qwen 3 0.6B and 1.7B achieve **nearly identical throughput** ($\sim$19.5 tok/s).
At this scale on T4, the bottleneck is **framework overhead**, not weight bandwidth.

Colab T4 — Decode Throughput per Prompt



Colab T4 — VRAM and TTFT





Qwen 3 0.6B: lowest VRAM (**1.24 GB**) and fastest TTFT (**54.7 ms**).

Qwen 3 — Thinking Mode ON vs OFF

Model	Thinking	TTFT (ms)	tok/s	Output tokens
Qwen 3 0.6B	OFF	54.7	19.6	134
Qwen 3 0.6B	ON	54.1	19.8	256 (capped)
Qwen 3 1.7B	OFF	55.2	19.4	136
Qwen 3 1.7B	ON	55.2	19.9	256 (capped)

Takeaway

Per-token speed is **unchanged**. Thinking mode generates $\sim 2\times$ the tokens (reasoning trace before answer) $\Rightarrow$ $\sim 2\times$ wall-clock time.
For latency-sensitive edge: leave thinking OFF.

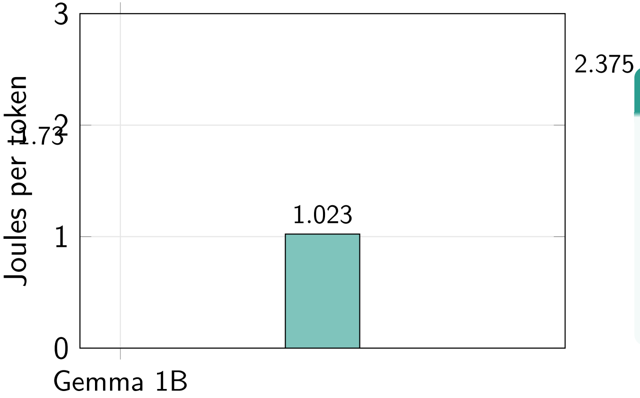
Jetson Xavier NX — Summary (Thinking OFF)

Model	**TTFT** (ms)	**tok/s**	**VRAM** (GB)	**PPL** ↓	**Power** (W)	**J/tok** ↓
Gemma 3 1B	279	5.2	2.05	26.57	9.0	1.730
Qwen 3 0.6B	193	**7.8**	**1.28**	21.23	8.0	**1.023**
Qwen 3 1.7B	247	4.2	3.54	**17.03**	9.8	2.375

Critical difference from T4

On T4: Qwen 0.6B ≈ Qwen 1.7B speed (19.5 tok/s).
On NX: Qwen 0.6B is **86% faster** than 1.7B (7.8 vs 4.2 tok/s).
⇒ NX is **bandwidth-bound**; parameter count matters on the edge.

Energy per Token — Xavier NX



Energy Winner: Qwen 3 0.6B

1.023 J/tok

41% less than Gemma 3 1B
57% less than Qwen 3 1.7B

For battery-powered edge, this is the deciding metric.

Full Cross-Platform Comparison

Model	Plat.	TTFT	tok/s	VRAM	PPL	Load	Power	J/tok
Gemma 3 1B	T4	67.4	15.9	2.02	26.57	26.1	—	—
	NX	279	5.2	2.05	26.57	42.8	9.0	1.730
Qwen 3 0.6B	T4	54.7	19.6	1.24	21.23	37.7	—	—
	NX	193	**7.8**	1.28	21.23	28.3	8.0	**1.023**
Qwen 3 1.7B	T4	55.2	19.4	3.50	17.02	177.6	—	—
	NX	247	4.2	3.54	**17.03**	65.4	9.8	2.375

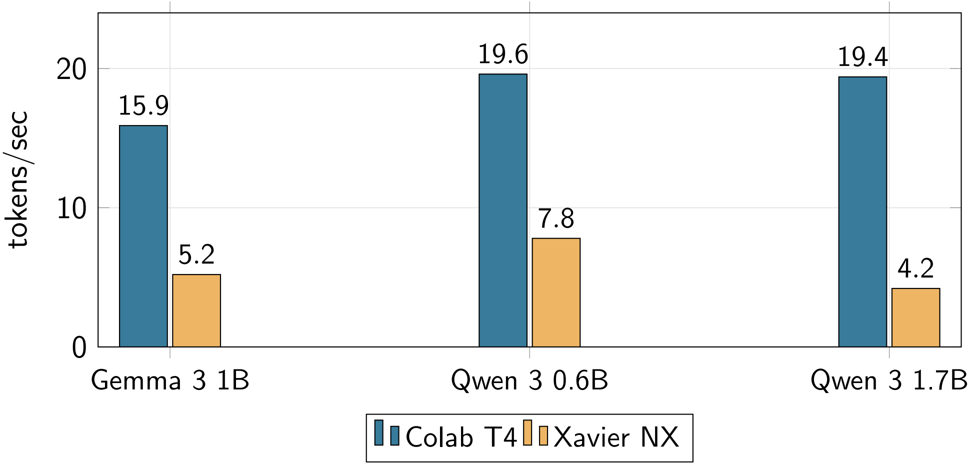
Speedup Ratios — T4 over Xavier NX

Model	T4 (tok/s)	NX (tok/s)	Ratio	Why
Gemma 3 1B	15.9	5.2	**3.06**×	SWA kernel slow on NX
Qwen 3 0.6B	19.6	7.8	**2.51**×	Least bandwidth-sensitive
Qwen 3 1.7B	19.4	4.2	**4.64**×	Most bandwidth-hungry

Analysis

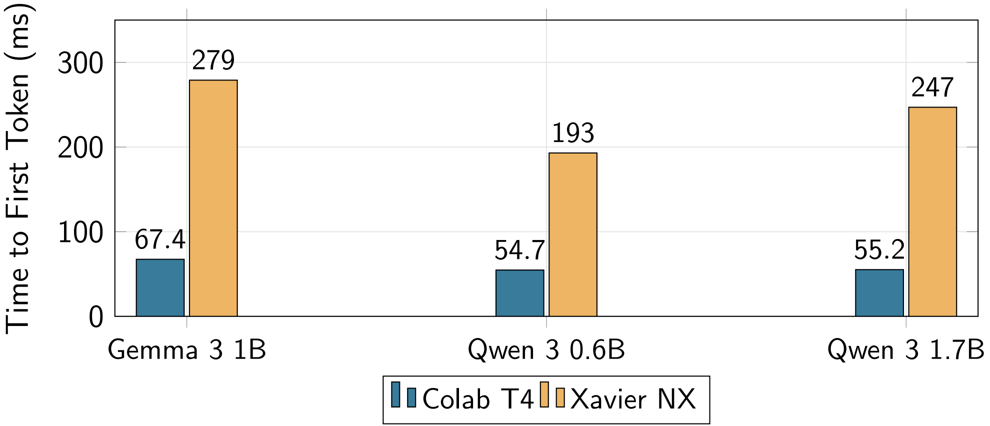
- Smaller models show **lower speedup ratios** — less bandwidth-sensitive
- Qwen 1.7B suffers the most: 3× more weights to move through 6.25× less bandwidth
- This confirms: **quantisation would disproportionately help NX**

Throughput Comparison — T4 vs Xavier NX



On T4: Qwen 0.6B $\approx$ 1.7B. On NX: Qwen 0.6B is **86% faster**. *The overhead ceiling breaks on the edge.*

TTFT Comparison — T4 vs Xavier NX



NX TTFT: 193–279 ms. Still sub-300 ms for all models — acceptable for interactive use.

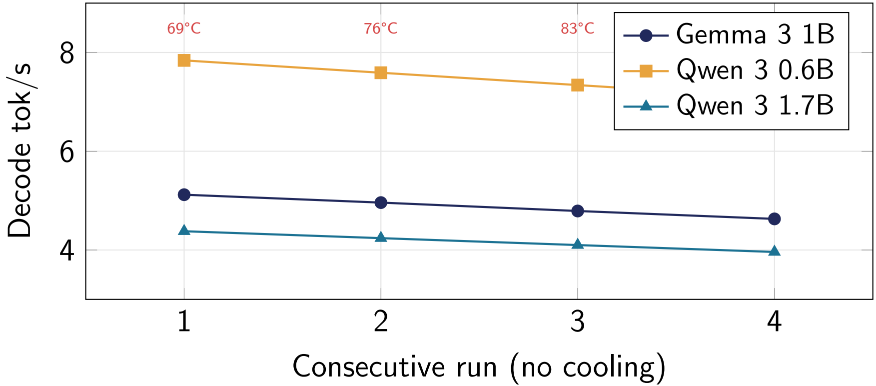
Perplexity — Cross-Platform Validation

Model	PPL (T4)	PPL (NX)	Δ
Gemma 3 1B	26.569	26.574	+0.02%
Qwen 3 0.6B	21.234	21.228	–0.03%
Qwen 3 1.7B	17.020	17.025	+0.03%

Numerical reproducibility confirmed

Same model, same weights, same fp16/SDPA $\Rightarrow$ PPL matches within **0.03%** across x86/Turing and ARM/Volta. The comparison methodology is sound.

Thermal Soak — 4 Consecutive Runs on Xavier NX



–9.6% throughput loss over 4 runs. GPU: 69°C → 90°C. NX's compact heatsink throttles under sustained load.

Thermal Implications for Deployment

Problem

- NX passive cooling insufficient for sustained inference
- 9.6% throughput loss after 4 back-to-back runs
- GPU hits 90°C — near throttle ceiling

Solutions

- Active cooling (fan module: $15–25)
- Duty-cycling inference with cooldown periods
- 10W mode (`nvpmodel -m 1`) — trades speed for thermal headroom
- Budget 10% throughput margin in SLAs

Finding 1: The Overhead Ceiling Breaks on the Edge

On Colab T4

Qwen 0.6B: **19.6** tok/s
Qwen 1.7B: **19.4** tok/s

$\Delta = 1\%$ — *effectively identical*

$\Rightarrow$ **Overhead-bound**: Python framework cost dominates, not weight transfer

On Xavier NX

Qwen 0.6B: **7.8** tok/s
Qwen 1.7B: **4.2** tok/s

$\Delta = 86\%$ — *massive gap*

$\Rightarrow$ **Bandwidth-bound**: 51.2 GB/s exposes weight-transfer cost

Implication: Deployment recommendations must be *platform-specific*.
Choosing 1.7B over 0.6B costs **nothing** on T4 but **86% throughput** on NX.

Finding 2: The Quality–Energy Trade-off

	PPL ↓	**J/tok ↓**	**tok/s ↑**	**Quality gain**	**Energy cost**
Qwen 0.6B	21.23	1.023	7.8	baseline	baseline
Qwen 1.7B	17.02	2.375	4.2	–20% PPL	+132% energy

Interpretation

Moving from 0.6B to 1.7B delivers **20% better perplexity** at **2.3× the energy cost**.

- Battery-powered / thermally constrained ⇒ **Qwen 3 0.6B**
- Tethered with active cooling ⇒ **Qwen 3 1.7B** (if 3.5 GB VRAM is available)

Finding 3: Gemma 3 1B is Pareto-Dominated

	tok/s	TTFT	VRAM	Disk	PPL	J/tok
Gemma 3 1B	5.2	279 ms	2.05	2.03	26.57	1.730
Qwen 3 0.6B	**7.8**	**193 ms**	**1.28**	**1.52**	**21.23**	**1.023**
Winner	Qwen	Qwen	Qwen	Qwen	Qwen	Qwen

Verdict

Qwen 3 0.6B beats Gemma 3 1B on **every single metric** on Xavier NX.
Despite having 25% fewer parameters and a smaller disk footprint.
Recommendation: do not deploy Gemma 3 1B on Jetson.

Deployment Recommendations

Memory-Constrained ($\leq$ 8 GB)

Qwen 3 0.6B

7.8 tok/s • 1.02 J/tok
1.28 GB VRAM • PPL 21.2

Best speed, best energy, smallest footprint

Quality-Priority ($\geq$ 16 GB)

Qwen 3 1.7B

4.2 tok/s • 2.38 J/tok
3.54 GB VRAM • PPL 17.0

20% better quality, needs active cooling

Avoid on Jetson

Gemma 3 1B — Pareto-dominated by Qwen 3 0.6B on all metrics

Future Work

1. **llama.cpp / TensorRT-LLM**: fused kernels expected to give 2–3× further speedup on NX
2. **Q4_K_M GGUF quantisation**: reduces bandwidth pressure, should narrow T4/NX gap
3. **10W power mode**: trade throughput for thermal sustainability
4. **IFEval / MMLU subset**: move beyond PPL to task-specific quality evaluation
5. **Orin Nano comparison**: next-gen Jetson with higher bandwidth

Thank You

Questions?

Code, data, and notebooks available at:
`[https://github.com/ArjunAmbekar-iitdh/ultralight-llm-evaluation.git]`